Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.2701713620336, median 0.24185096275595297, std: 0.1639390988293921
Gyroscope error (imu0): mean 0.04931406553795156, median 0.03194087221500021, std: 0.08602501945296308
Accelerometer error (imu0): mean 0.06481675756241005, median 0.046138206475095346, std: 0.10501007353161464

Residuals

Reprojection error (cam0) [px]: mean 0.2701713620336, median 0.24185096275595297, std: 0.1639390988293921
Gyroscope error (imu0) [rad/s]: mean 0.0019471237609418651, median 0.0012611580602124926, std: 0.0033966244231768594
Accelerometer error (imu0) [m/s^2]: mean 0.025164971641315908, median 0.017913062936062443, std: 0.04076994317296922

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.03202743 -0.99651157 0.0770645 -0.0066671]
[0.09374535 -0.07975918 -0.99239623 0.02523279]
[0.99508093 -0.02455946 0.09597281 -0.0311839]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[-0.03202743 0.09374535 0.99508093 0.02845152]
[-0.99651157 -0.07975918 -0.02455946 -0.00539716]
[0.0770645 -0.99239623 0.09597281 0.02854753]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.00031673965517683766

Gravity vector in target coords: [m/s^2]
[0.14367798 -9.80458514 -0.13375231]

Calibration configuration
==========

cam0

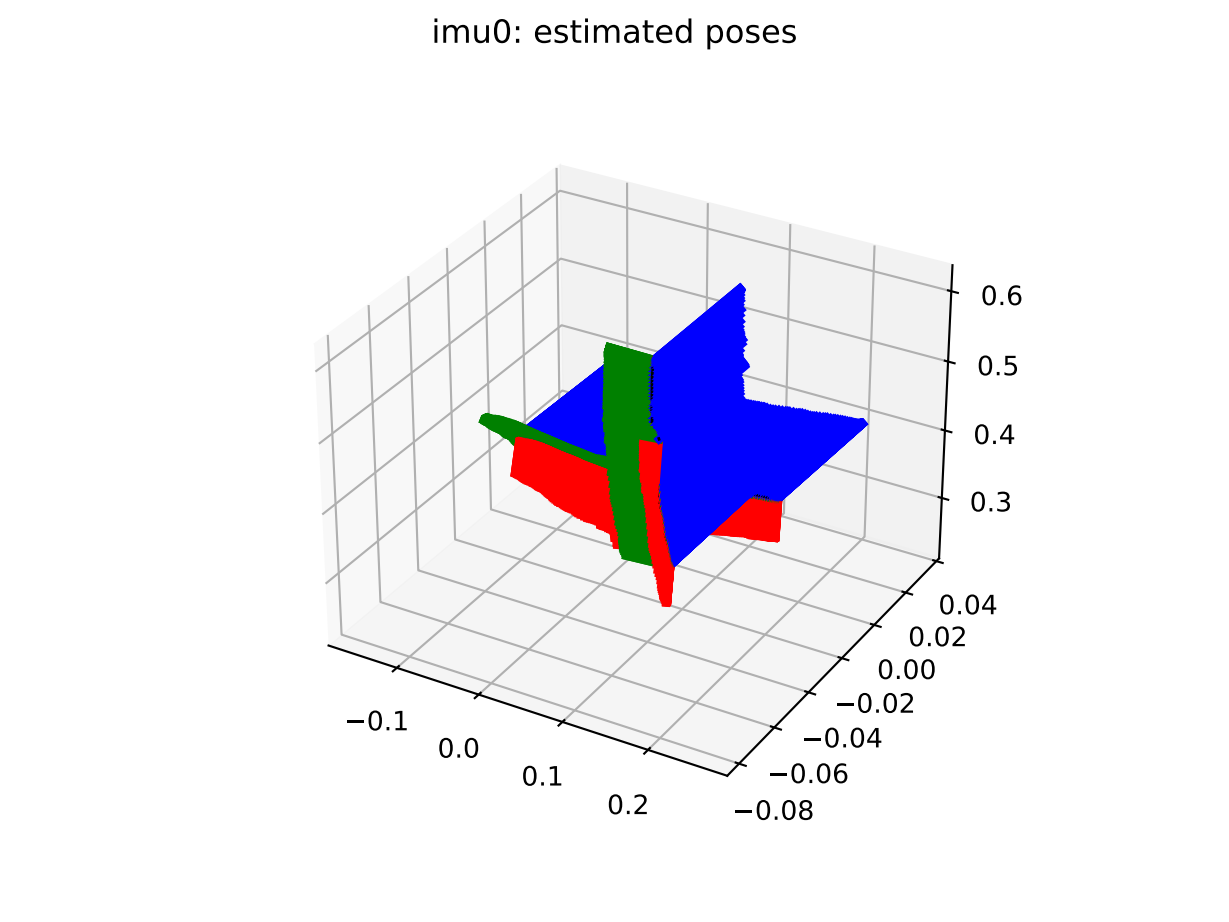
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.02 [m]
    Spacing 0.006 [m]

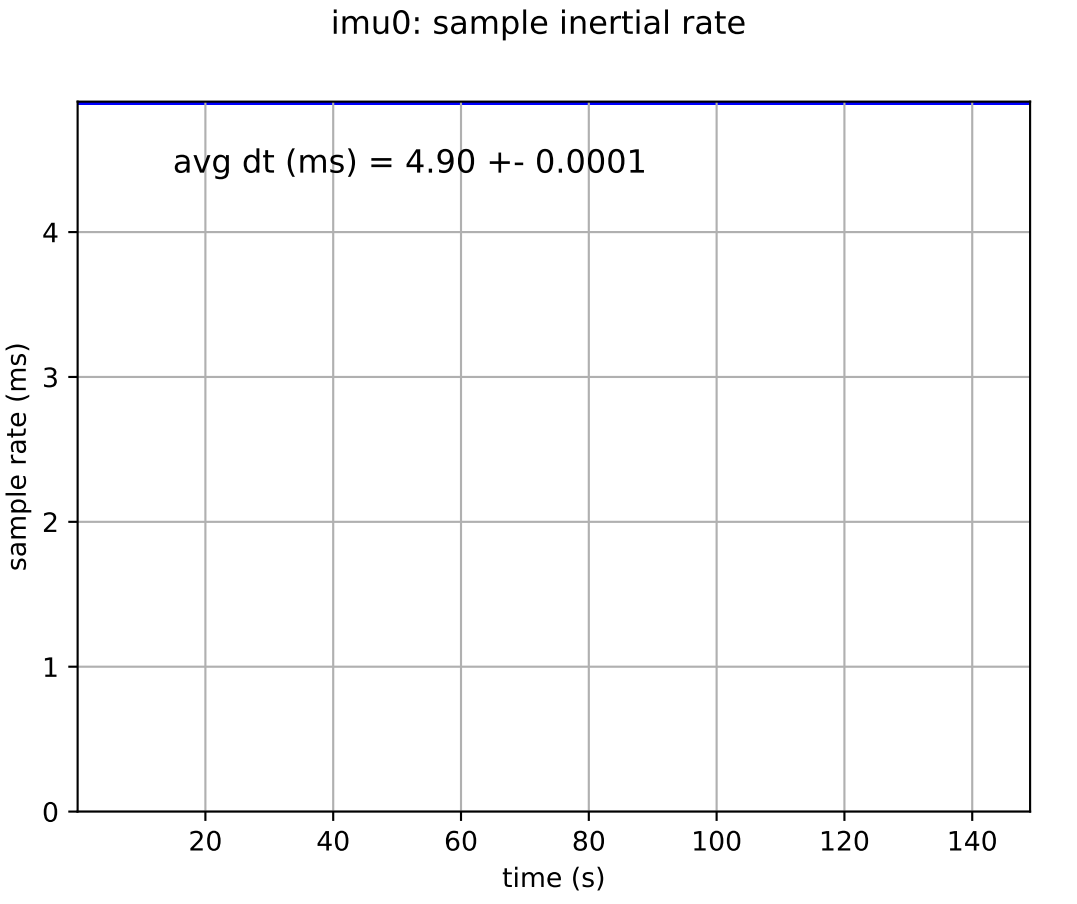
IMU configuration
==========

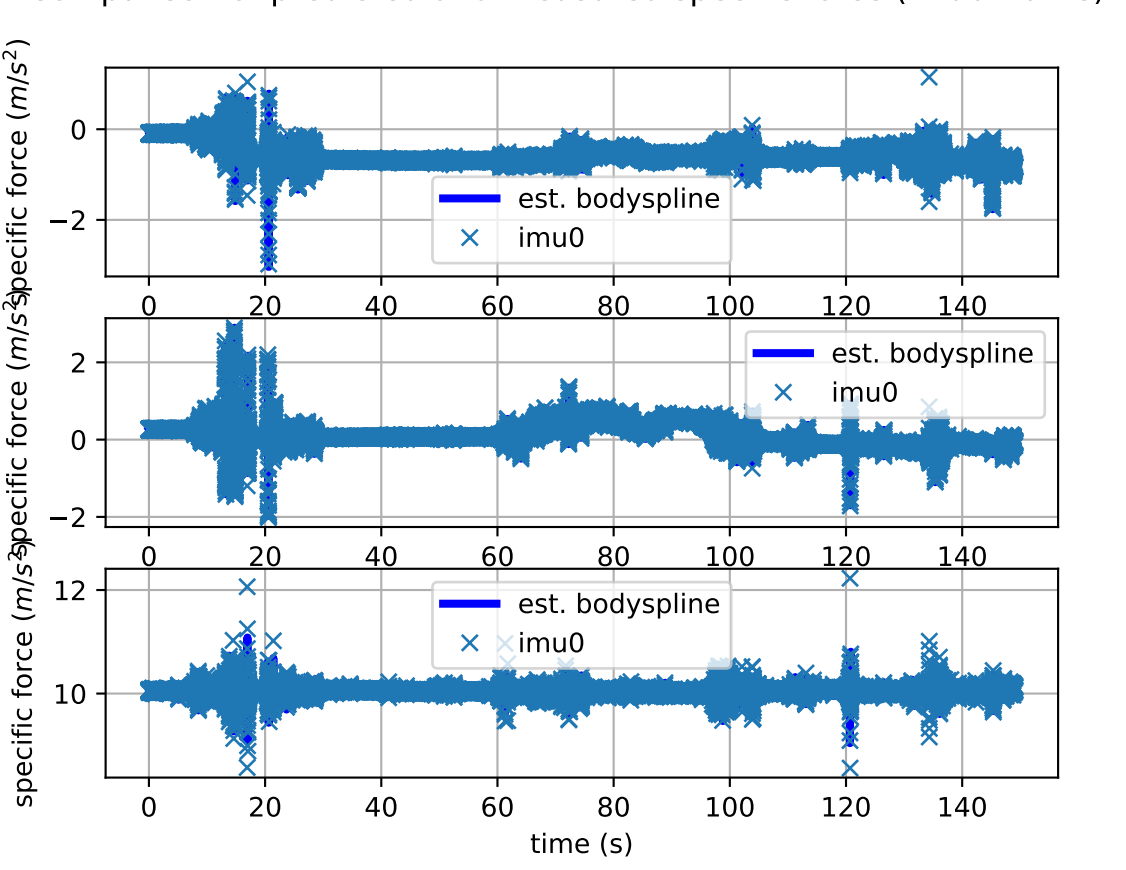
IMU0:

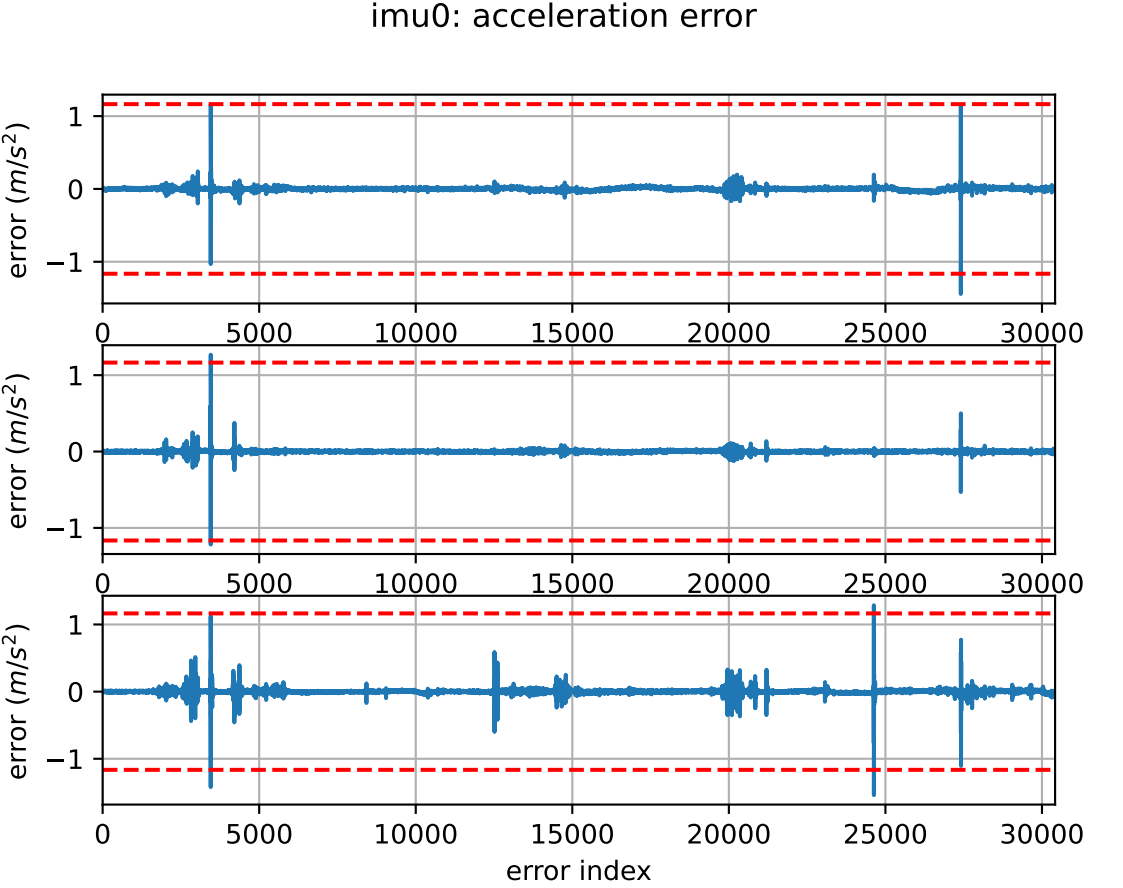
  Model: calibrated
  Update rate: 204.090470950253
  Accelerometer:
    Noise density: 0.027176766288825538
    Noise density (discrete): 0.38824792519257595
    Random walk: 0.0067023621585376265
  Gyroscope:
    Noise density: 0.002763830369432253
    Noise density (discrete): 0.039484145947029654
    Random walk: 0.00014770260860981916

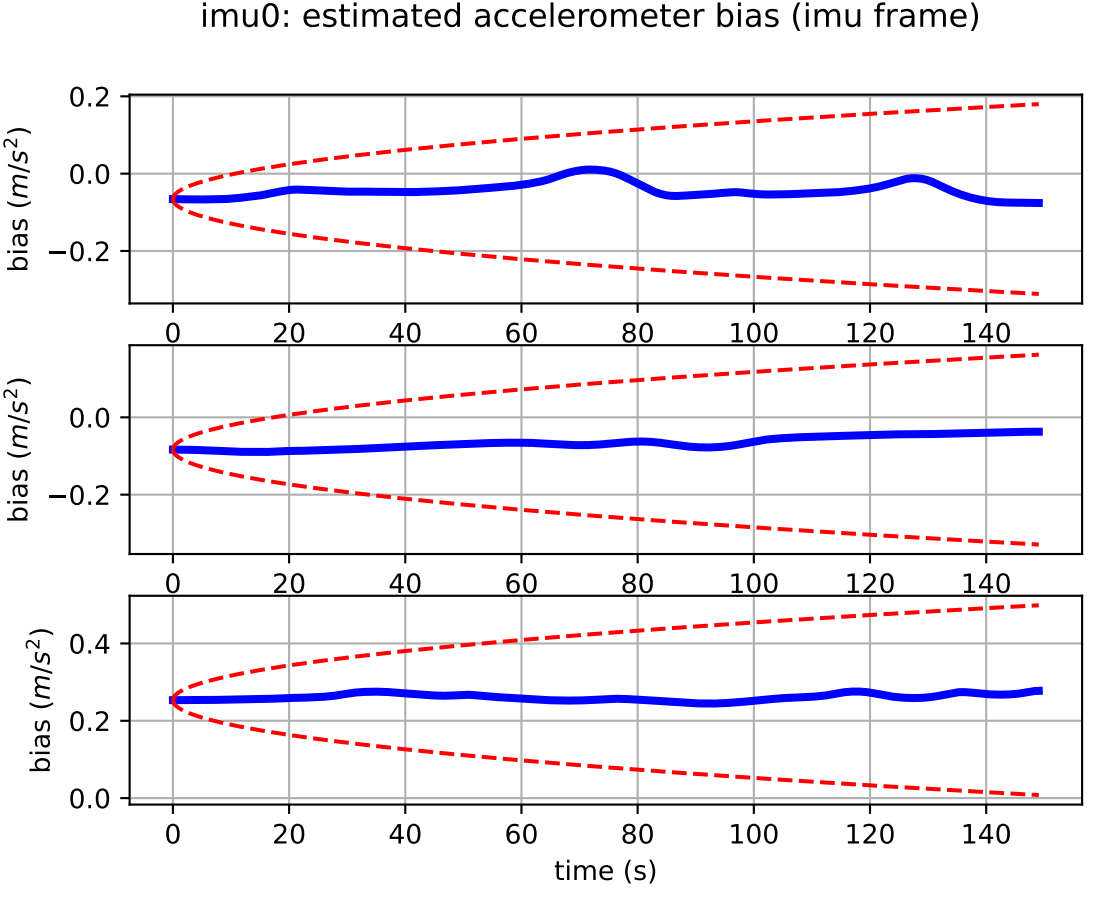
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

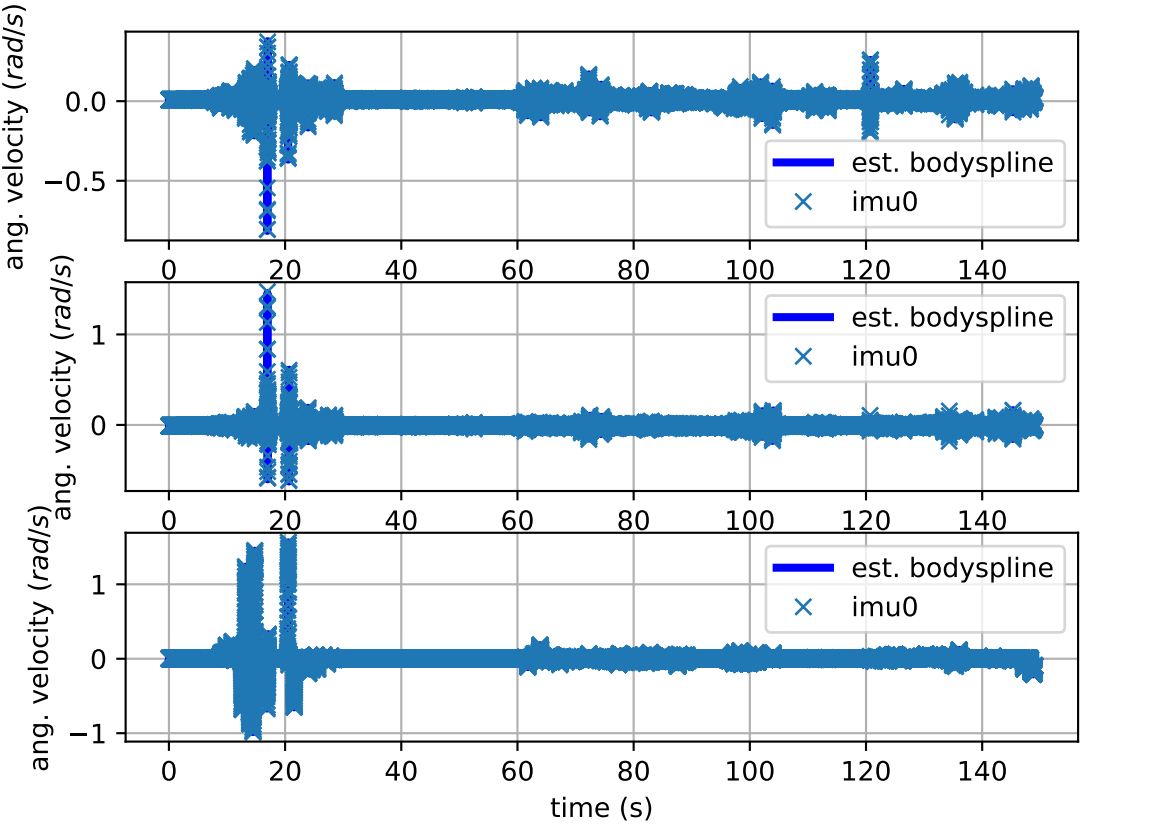

imu0: estimated poses
0.6
0.5
0.4
0.3
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−0.1
0.0
0.1
0.2

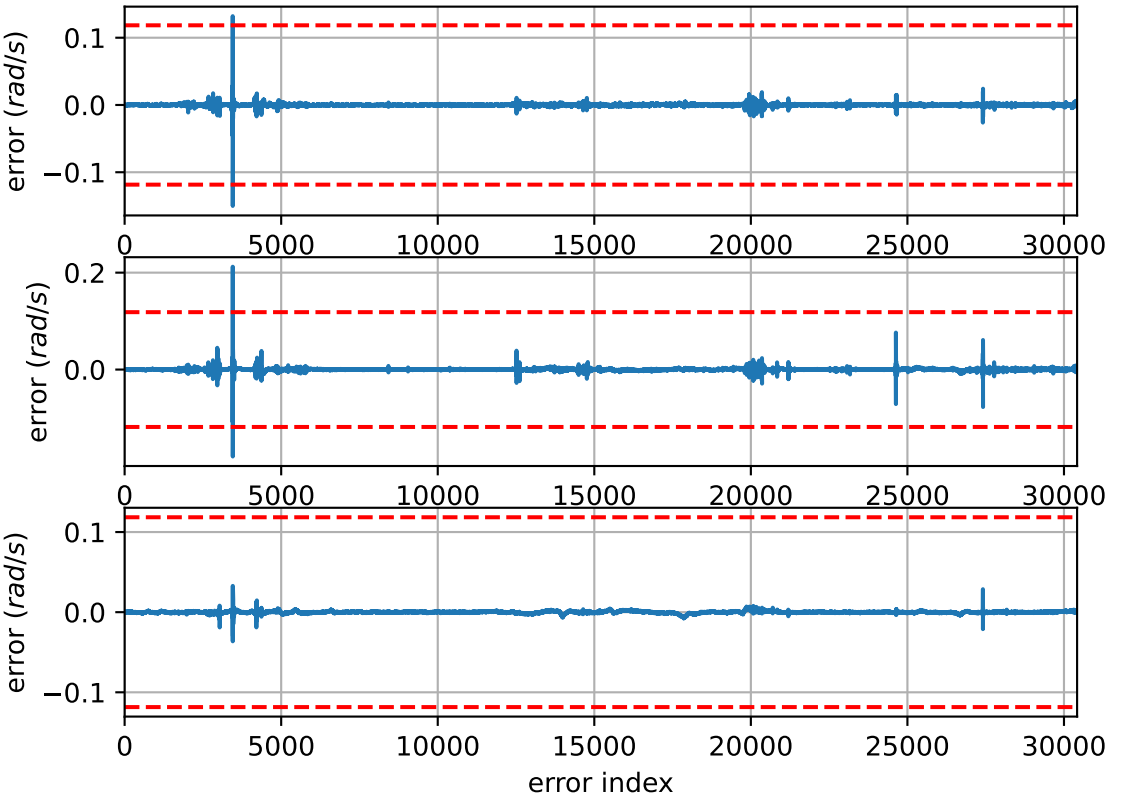

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0001
sample rate (ms)
0
1
2
3
4
20
40
60
80
100
120
140
time (s)

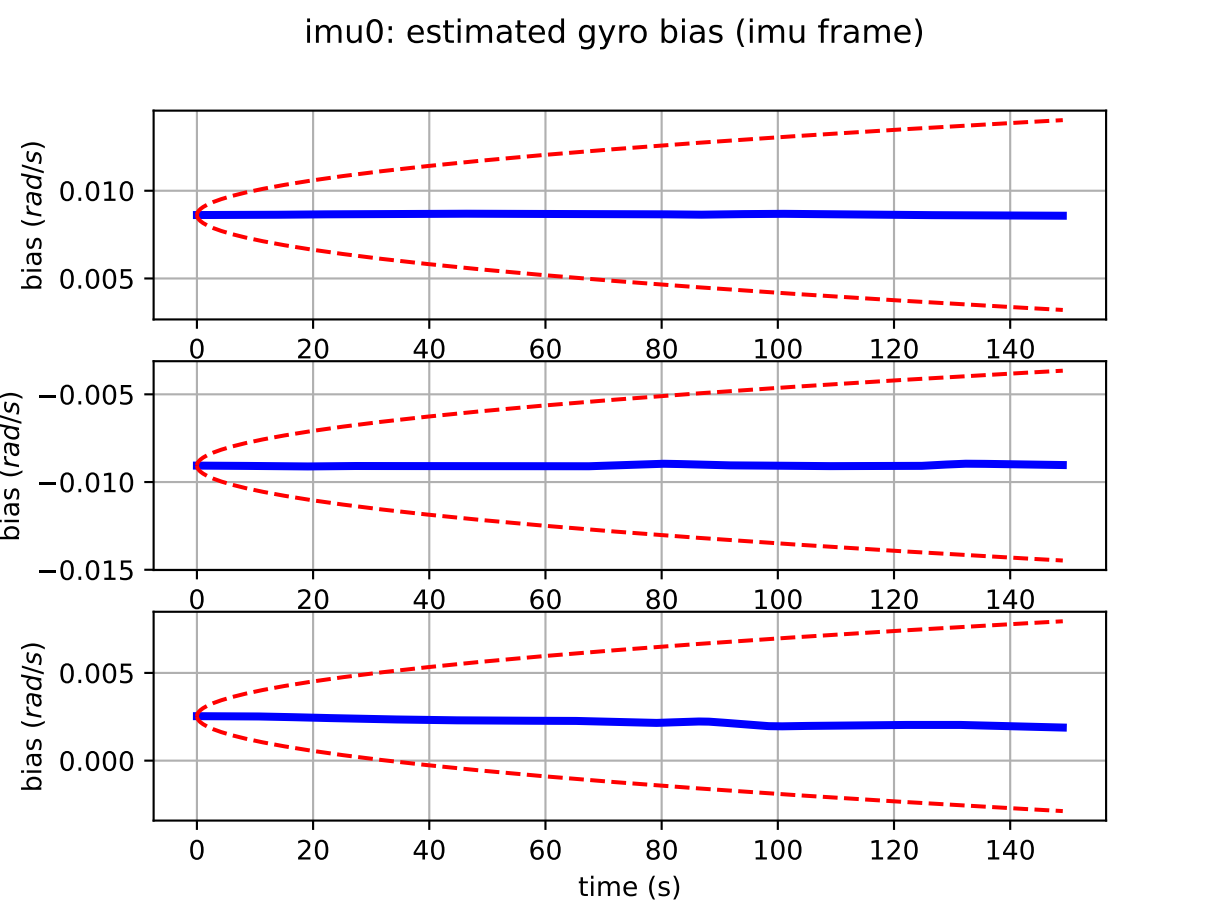

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
time (s)

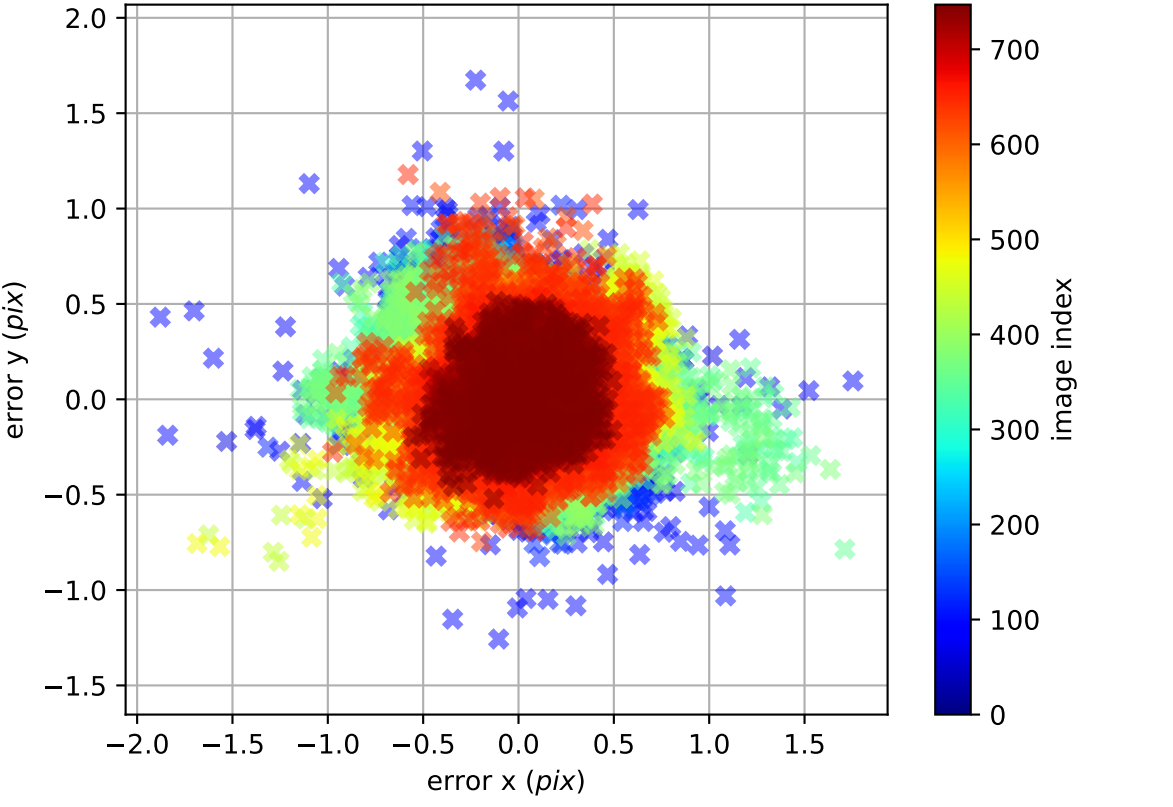

imu0: acceleration error
error (m/s^2)
1
0
−1
0
5000
10000
15000
20000
25000
30000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.2
0.0
−0.2
0
20
40
60
80
100
120
140
bias (m/s^2)
0.0
−0.2
0
20
40
60
80
100
120
140
bias (m/s^2)
0.4
0.2
0.0
0
20
40
60
80
100
120
140
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
0.0
−0.5
est. bodyspline
imu0
0
20
40
60
80
100
120
140
ang. velocity (rad/s)
1
0
est. bodyspline
imu0
ang. velocity (rad/s)
1
0
−1
est. bodyspline
imu0
time (s)


imu0: angular velocities error
error (rad/s)
0.1
0.0
−0.1
0
5000
10000
15000
20000
25000
30000
error (rad/s)
0.2
0.0
0
5000
10000
15000
20000
25000
30000
error (rad/s)
0.1
0.0
−0.1
0
5000
10000
15000
20000
25000
30000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.010
0.005
0
20
40
60
80
100
120
140
bias (rad/s)
−0.005
−0.010
−0.015
bias (rad/s)
0.005
0.000
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
700
600
500
400
300
200
100
0